



AHB Example Bus Master Technical Reference Manual

IP Solutions Division

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Abstract

The example bus master provides a simple demonstration of an AMBA AHB bus master. The component is comprised of two blocks, a simple AHB-Lite master which generates a fixed sequence of transfers and an AHB-Lite to AHB wrapper which enables its connection to a standard AHB bus. This is intended to act as an example of both AHB-Lite bus master design and the use of the AHB-Lite to AHB wrapper.

Keywords

Example AHB Bus Master AHB-Lite to AHB Wrapper AMBA Bus

Distribution list

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1 ABOUT THIS DOCUMENT

1.1 Change control

1.1.1 Current status and anticipated changes

1.1.2 Change history

Issue	Date	By	Change
A00	13 March, 2000	Antony Harris	First draft
A01	20 April, 2000	Antony Harris	Added MLOCK to Mbus, updated block names
A02	3 August, 2000	Antony Harris	Mostly rewritten
A03	5 October, 2000	Antony Harris	MBUSREQ removed

1.2 References

This document refers to the following documents.

Ref	Doc No	Author(s)	Title
R1	ARM IHI 0011A	B Mathewson	AMBA Specification
R2	AMBA DISC 0007 A03	B Mathewson	AHB-Lite Specification
R3	ARM DDI 0170A	B Mathewson	Example AMBA SYstem TRM

1.3 Terms and abbreviations

This document uses the following terms and abbreviations.

Term	Meaning
AMBA	Advanced Microcontroller Bus Architecture
AHB	Advanced High-performance Bus
AHB-Lite	Simplified AHB specification for single master systems
EBM	Example Bus Master

2 SCOPE

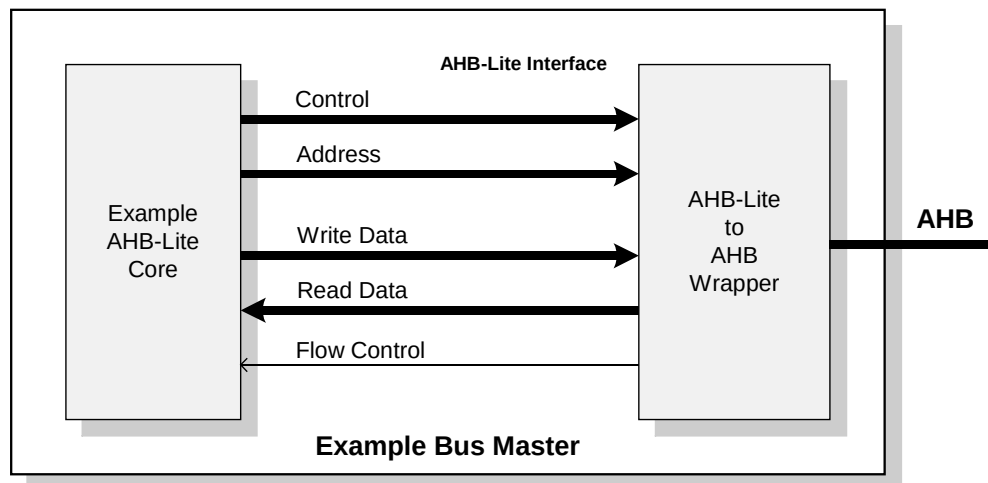
This document provides a reference specification for both the AHB example bus master and AHB-Lite to AHB wrapper.

3 INTRODUCTION

AHB-Lite is a subset of the full AHB specification and is intended for use in designs where only a single bus master is used. This may either be a simple single master system or a Multi-layer AHB system where there is only one AHB master per layer.

AHB-Lite simplifies the AHB specification by removing the protocol required for multiple bus masters, which includes the Request/Grant protocol to the Arbiter and the Split/Retry responses from slaves. Masters designed to the AHB-Lite interface specification can be significantly simpler in terms of interface design, compared to a full AHB master. This allows faster design and verification of such masters and the addition of a standard “off-the-shelf” bus mastering wrapper can be used to convert an AHB-Lite master for use in a full AHB system.

The example bus master consists of an AHB-Lite core, which generates a fixed set of transfers in a continuous fashion and an AHB-Lite to AHB wrapper, which enables its connection to a standard AHB bus. This is intended to provide an example of both AHB-Lite bus master design and the use of the wrapper to interface to an AHB system. This can also be used in the EASY system, in place of an ARM processor model, to enable rapid simulation of AHB transactions.



4 EXAMPLE AHB-LITE CORE

The example AHB-Lite core consists of a counter, burst generation logic and a 4 x 32 bit register bank. From a *power-up* or reset state, the core uses the counter to wait for a predefined number of clock cycles. It then performs a 4-beat, incrementing, word, read burst from a parameterised base source address into the register bank. Once the burst completes, the bus master waits for further predefined time, before writing data from the registers as 16 single byte transfers. This data is written, least significant byte first, to a single non-incrementing destination address (such as the EASY Tube peripheral). This process continues, using the same source and destination addresses until a *power-down* or reset condition is reached.

The operation of the bus master is fixed, but a number of parameters may be changed at compile-time. These are the number of clock cycles for the delays (10 bits), the base source address and the destination address (both word aligned).

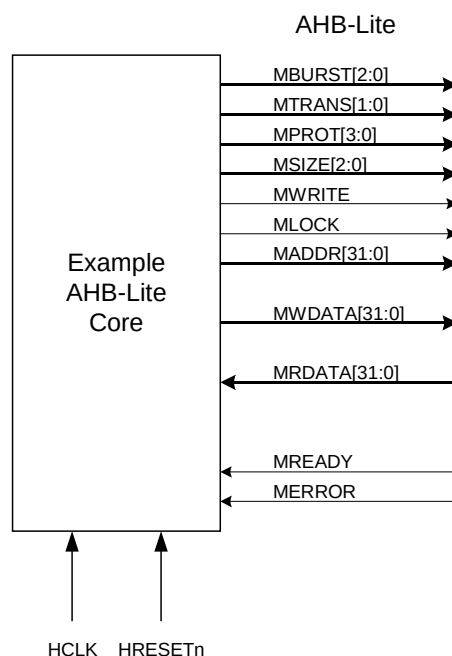


Figure 1

The schematic for the example core is shown in Figure 1 where the signals on the AHB-Lite bus have been prefixed with 'M' in order to differentiate them from the AHB signals, prefixed with 'H'.

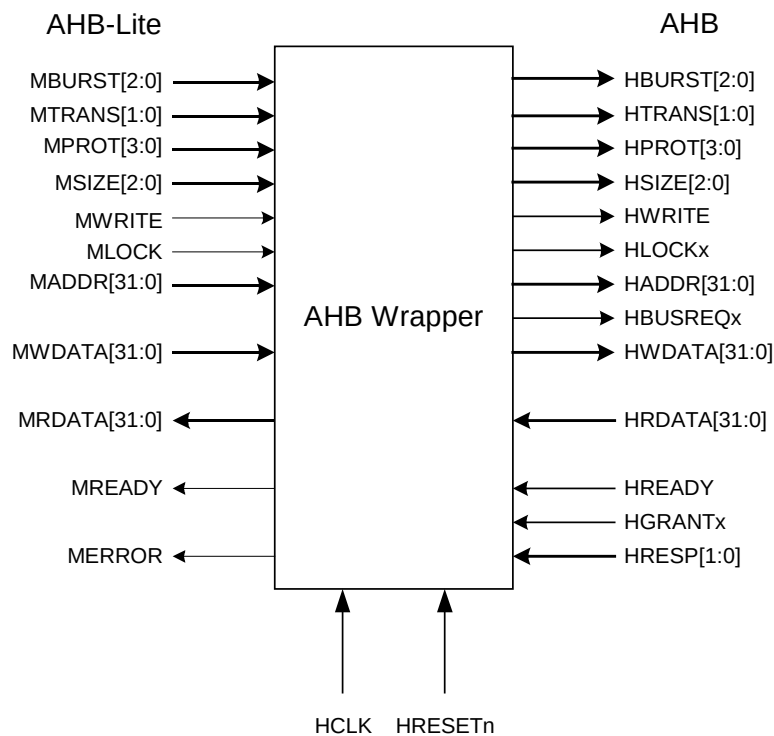
The bus master is completely synchronous with the AHB bus clock signal, HCLK and is reset by the AHB reset signal HRESETn. The example core does not use locked transfers, so the MLOCK signal is held low throughout the transfers. The MREADY signal is used to stall the core as per the AHB-Lite specification.

The signal MERROR actually represents signal HRESP[0] as SPLIT/RETRY responses are not supported in AHB-Lite. When this is asserted during a read operation, the corresponding data is ignored.

5 AHB-LITE TO AHB WRAPPER

The AHB-Lite to AHB wrapper is designed to enable any AHB-Lite master to connect to a standard AHB bus. This requires the wrapper to add support for the features of:

- Bus ownership – when a transfer is initiated by the master (MTRANS is NONSEQ), the wrapper generates a request using HBUSREQ. The bus is also re-requested if ownership is lost mid-burst.
- Early terminated bursts – if a burst is terminated prematurely through loss of the bus, then the wrapper holds the master using MREADY and rebuilds the burst when bus control is re-gained.
- Slave responses – the slave responses of Split and Retry are handled by the wrapper. Bursts are also regenerated in these cases. Error responses are passed to the master through the signal MERROR (equivalent to HRESP[0]) and it is left to the master to determine an appropriate action.



6 FUNCTIONAL DESCRIPTION

As an illustration of the functional interaction between the example AHB-Lite core and AHB-Lite to AHB wrapper, Figure 2 illustrates a read burst, the first transfer of which is responded to with a RETRY.

- T1. The transfer begins when the EBM drives MTRANS to NONSEQ, together with the address and other control signals. The wrapper asserts HBUSREQebm to request control of the AHB.
- T2. The EBM drives the address and control for the second transfer in the burst. The holding registers in the wrapper are used to keep HADDR at A1. HGRANTEbm goes high, but control of the address bus is not yet gained. HTRANS is driven to IDLE, as driving SEQ when not in control of the bus is not legal. MREADY is de-asserted as the data D1 is not yet available.
- T3. HGRANTEbm is sampled high, so HBUSREQebm is de-asserted. The first transfer is put onto the AHB, MREADY is held low.
- T4. The data D1 is driven by the slave and passes through the wrapper, MREADY is driven high to indicate that the data is available. The second transfer is placed on the bus.
- T5. The slave responds with RETRY to the A2 transfer. MREADY goes low to reflect HREADY.
- T6. HGRANTEbm is de-asserted due to the RETRY response, so HBUSREQebm is re-asserted. HTRANS is driven to IDLE in response to the RETRY. MREADY remains low to stall the core.
- T7. Control of the address bus is lost, so HTRANS is driven with IDLE. MREADY remains low.
- T8. The EBM has control of the address bus, so the transfer is re-attempted. As the core has moved on to the next transfer, the wrapper uses holding registers to drive the transfer signals. Note that HBURST must be modified for the remainder of the burst and HBUSREQdma is held high as the burst is now of unspecified
- T9. MREADY is asserted, releasing the core to complete the burst.

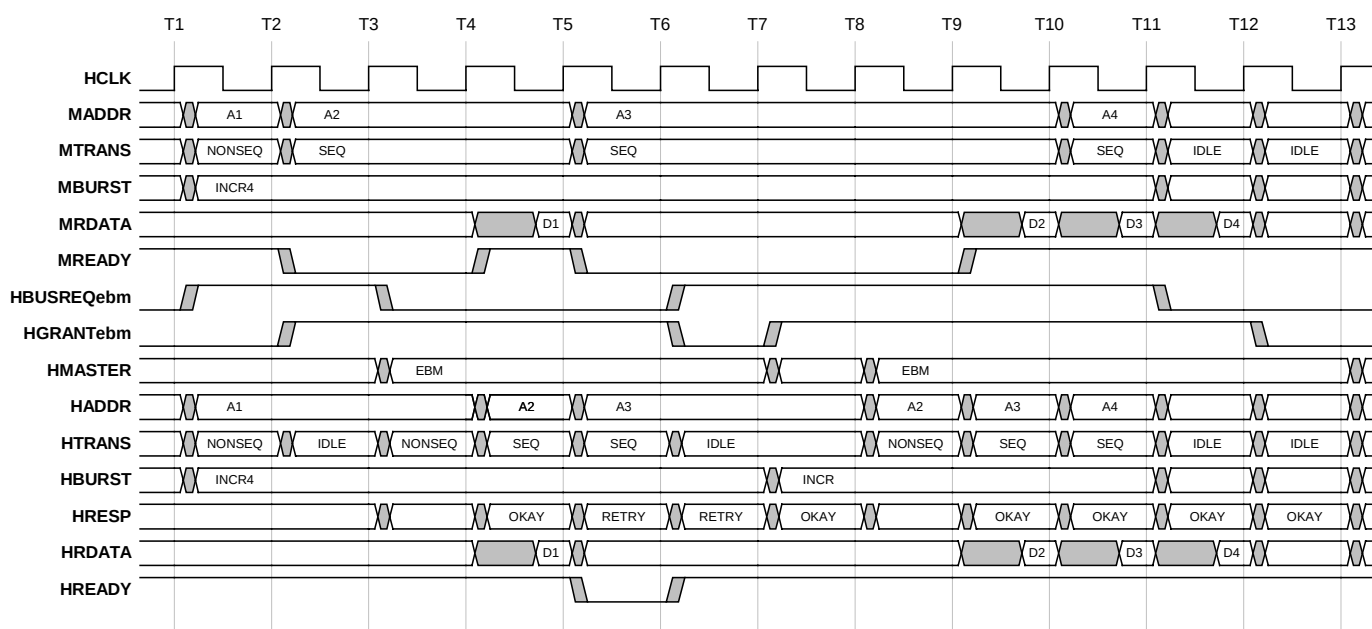
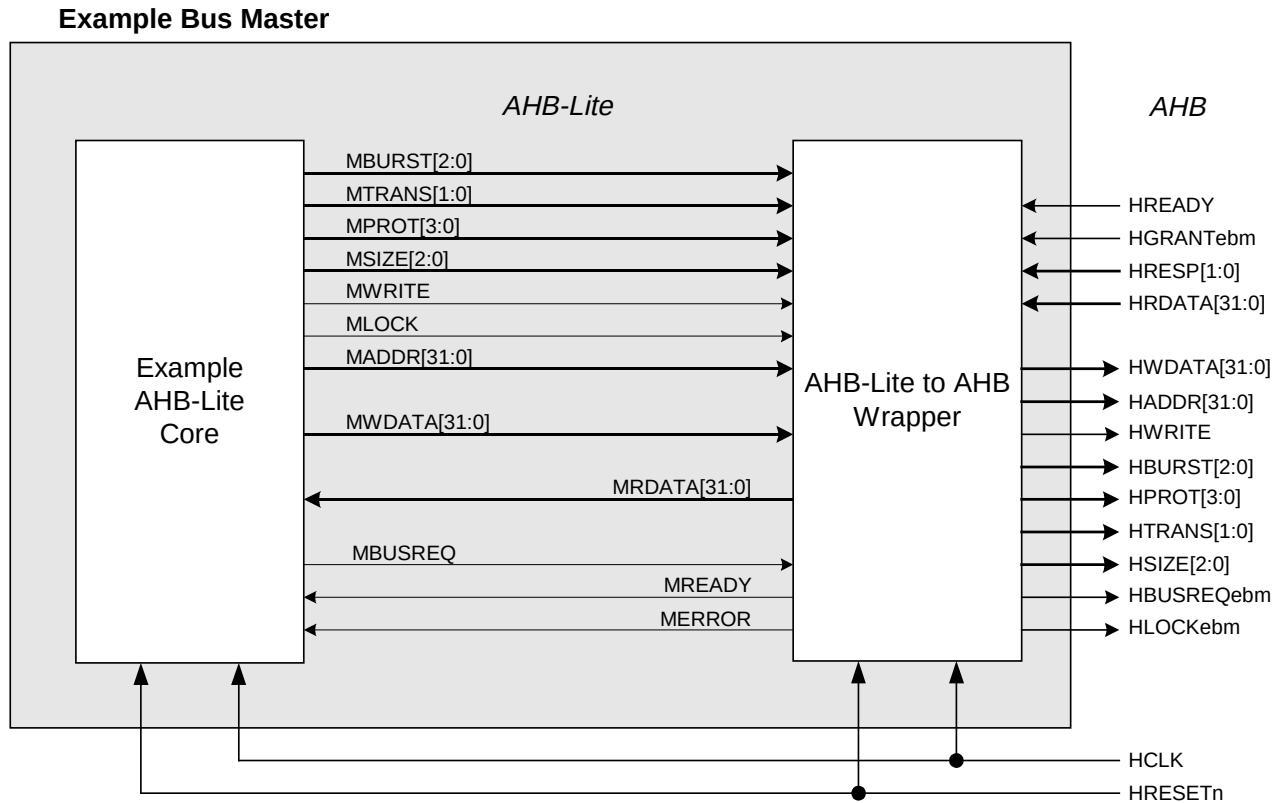


Figure 2 Read burst with retry response

7 SCHEMATIC DIAGRAM



8 INTERFACE SIGNALS

Name	Type	Description
HCLK Bus clock	Input	This clock times all bus transfers. All signal timings are related to the rising edge of HCLK .
HRESETn Reset	Input	The bus reset signal is active LOW and is used to reset the system and the bus.
HGRANTebm Bus grant	Input	This signal indicates that the example bus master is currently the highest priority master. Ownership of the address/control signals changes at the end of a transfer when HREADY is HIGH, so the master gets access to the bus when both HREADY and HGRANTebm are HIGH.
HREADY Transfer done	Input	When HIGH the HREADY signal indicates that a transfer has finished on the bus. This signal may be driven LOW to extend a transfer.
HRESP[1:0] Transfer response	Input	The transfer response provides additional information on the status of a transfer. Four different responses are provided, OKAY, ERROR, RETRY and SPLIT.
HRDATA[31:0] Read data bus	Input	The read data bus is used to transfer data from bus slaves to the bus master during read operations. This has a width of 32 bits for the example bus master.

Name	Type	Description
HBUSREQebm Bus request	Output	A signal from the example bus master to the arbiter, which indicates that the bus master requires the bus. There is an HBUSREQ signal for each bus master in the system.
HADDR[31:0] Address bus	Output	The 32-bit system address bus.
HTRANS[1:0] Transfer type	Output	Indicates the type of the current transfer, which can be NONSEQUENTIAL, SEQUENTIAL, IDLE or BUSY.
HWRITE Transfer direction	Output	When HIGH this signal indicates a write transfer and when LOW a read transfer.
HBURST[2:0] Burst type	Output	Indicates if the transfer forms part of a burst. The example bus master supports four beat incrementing bursts and single transfers.
HPROT[3:0] Protection control	Output	The protection control signals provide additional information about a bus access and are primarily intended for use by any module that wished to implement some level of protection. The example bus master uses a protection level of <i>data, user, not bufferable, not cacheable</i> .
HSIZE[2:0] Transfer size	Output	Indicates the size of the transfer. The example bus master uses 32 bit read and 8 bit write transfers.
HLOCKebm Locked transfers	Output	When HIGH, this signal indicates that the master requires locked access to the bus and no other master should be granted the bus until this signal is LOW. The example bus master does not utilise locked transfers.
HWDATA[31:0] Write data bus	Output	The write data bus is used to transfer data from the master to the bus slaves during write operations. This has a width of 32 bits for the example bus master.

9 AC CHARACTERISTICS

The example AHB bus master complies with the timing characteristics as defined in the AHB-Lite and AHB specifications.

10 MANUFACTURING TEST STRATEGY

The example bus master is designed for scan test insertion.